

Importing the libraries

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

plt.style.use('ggplot')
```

Loading both the datasets

```
trades = pd.read_csv("historical_data.csv")
fear = pd.read_csv("fear_greed_index.csv")
```

View the first 5 rows

```
print("Historical Data")
display(trades.head())

print("\nFear & Greed Data")
display(fear.head())
```

Historical Data

	Account	Coin	Execution Price	Size Tokens	Size USD	Side	Timestamp IST	Start Position	Direction	Closed PnL
0	0xae5eacaf9c6b9111fd53034a602c192a04e082ed	@107	7.9769	986.87	7872.16	BUY	02-12-2024 22:50	0.000000	Buy	0.0
1	0xae5eacaf9c6b9111fd53034a602c192a04e082ed	@107	7.9800	16.00	127.68	BUY	02-12-2024 22:50	986.524596	Buy	0.0
2	0xae5eacaf9c6b9111fd53034a602c192a04e082ed	@107	7.9855	144.09	1150.63	BUY	02-12-2024 22:50	1002.518996	Buy	0.0
3	0xae5eacaf9c6b9111fd53034a602c192a04e082ed	@107	7.9874	142.98	1142.04	BUY	02-12-2024 22:50	1146.558564	Buy	0.0
4	0xae5eacaf9c6b9111fd53034a602c192a04e082ed	@107	7.9894	8.73	69.75	BUY	02-12-2024 22:50	1289.488521	Buy	0.0

Fear & Greed Data

	timestamp	value	classification	date
0	1517463000	30	Fear	2018-02-01
1	1517549400	15	Extreme Fear	2018-02-02
2	1517635800	40	Fear	2018-02-03
3	1517722200	24	Extreme Fear	2018-02-04
4	1517808600	11	Extreme Fear	2018-02-05

Checking the dataset information

```
print("Historical Data Info")
trades.info()
```

```
print("\nFear & Greed Info")
fear.info()
```

```
Historical Data Info
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 211224 entries, 0 to 211223
Data columns (total 16 columns):
```

```

#   Column          Non-Null Count  Dtype
---  -
0   Account         211224 non-null    object
1   Coin             211224 non-null    object
2   Execution Price   211224 non-null    float64
3   Size Tokens      211224 non-null    float64
4   Size USD         211224 non-null    float64
5   Side             211224 non-null    object
6   Timestamp IST     211224 non-null    object
7   Start Position    211224 non-null    float64
8   Direction        211224 non-null    object
9   Closed PnL       211224 non-null    float64
10  Transaction Hash  211224 non-null    object
11  Order ID         211224 non-null    int64
12  Crossed          211224 non-null    bool
13  Fee              211224 non-null    float64
14  Trade ID         211224 non-null    float64
15  Timestamp         211224 non-null    float64
dtypes: bool(1), float64(8), int64(1), object(6)
memory usage: 24.4+ MB

```

```

Fear & Greed Info
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2644 entries, 0 to 2643
Data columns (total 4 columns):
#   Column          Non-Null Count  Dtype
---  -
0   timestamp       2644 non-null    int64
1   value           2644 non-null    int64
2   classification  2644 non-null    object
3   date            2644 non-null    object
dtypes: int64(2), object(2)
memory usage: 82.8+ KB

```

```

# Convert Timestamp IST (automatically detect mixed formats)
trades['Timestamp IST'] = pd.to_datetime(
    trades['Timestamp IST'],
    format='mixed',
    dayfirst=True
)

# Convert Fear & Greed date
fear['date'] = pd.to_datetime(fear['date'])

# Create Date column
trades['Date'] = trades['Timestamp IST'].dt.date
fear['Date'] = fear['date'].dt.date

```

Merge the datasets

```

merged_df = pd.merge(
    trades,
    fear[['Date', 'classification', 'value']],
    on='Date',
    how='left'
)

print("Merged Dataset Shape:", merged_df.shape)
merged_df.head()

```

Merged Dataset Shape: (211224, 19)

	Account	Coin	Execution Price	Size Tokens	Size USD	Side	Timestamp IST	Start Position	Direction	Closed PnL
0	0xae5eacaf9c6b9111fd53034a602c192a04e082ed	@107	7.9769	986.87	7872.16	BUY	2024-12-02 22:50:00	0.000000	Buy	0.0
1	0xae5eacaf9c6b9111fd53034a602c192a04e082ed	@107	7.9800	16.00	127.68	BUY	2024-12-02 22:50:00	986.524596	Buy	0.0
2	0xae5eacaf9c6b9111fd53034a602c192a04e082ed	@107	7.9855	144.09	1150.63	BUY	2024-12-02 22:50:00	1002.518996	Buy	0.0
3	0xae5eacaf9c6b9111fd53034a602c192a04e082ed	@107	7.9874	142.98	1142.04	BUY	2024-12-02 22:50:00	1146.558564	Buy	0.0
4	0xae5eacaf9c6b9111fd53034a602c192a04e082ed	@107	7.9894	8.73	69.75	BUY	2024-12-02 22:50:00	1289.488521	Buy	0.0

merged_df[['Date', 'Coin', 'Closed PnL', 'classification', 'value']].head(10)

	Date	Coin	Closed PnL	classification	value	
0	2024-12-02	@107	0.0	Extreme Greed	80.0	
1	2024-12-02	@107	0.0	Extreme Greed	80.0	
2	2024-12-02	@107	0.0	Extreme Greed	80.0	
3	2024-12-02	@107	0.0	Extreme Greed	80.0	
4	2024-12-02	@107	0.0	Extreme Greed	80.0	
5	2024-12-02	@107	0.0	Extreme Greed	80.0	
6	2024-12-02	@107	0.0	Extreme Greed	80.0	
7	2024-12-02	@107	0.0	Extreme Greed	80.0	
8	2024-12-02	@107	0.0	Extreme Greed	80.0	
9	2024-12-02	@107	0.0	Extreme Greed	80.0	

Check for Mising Values

merged_df.isnull().sum()

	0
Account	0
Coin	0
Execution Price	0
Size Tokens	0
Size USD	0
Side	0
Timestamp IST	0
Start Position	0
Direction	0
Closed PnL	0
Transaction Hash	0
Order ID	0
Crossed	0
Fee	0
Trade ID	0
Timestamp	0
Date	0
classification	6
value	6

dtype: int64

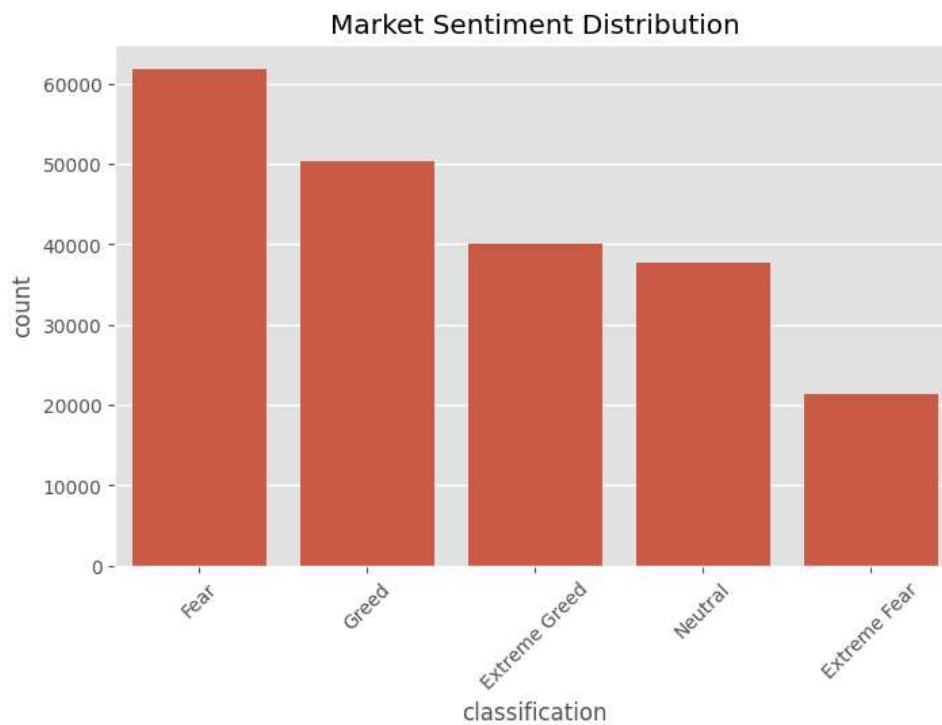
merged_df.describe()

	Execution Price	Size Tokens	Size USD	Timestamp IST	Start Position	Closed PnL	Order ID	Fee	
count	211224.000000	2.112240e+05	2.112240e+05	211224	2.112240e+05	211224.000000	2.112240e+05	211224.000000	2.
mean	11414.723350	4.623365e+03	5.639451e+03	2025-01-31 12:04:22.915009792	-2.994625e+04	48.749001	6.965388e+10	1.163967	5.
min	0.000005	8.740000e-07	0.000000e+00	2023-05-01 01:06:00	-1.433463e+07	-117990.104100	1.732711e+08	-1.175712	0.
25%	4.854700	2.940000e+00	1.937900e+02	2024-12-31 21:00:45	-3.762311e+02	0.000000	5.983853e+10	0.016121	2.
50%	18.280000	3.200000e+01	5.970450e+02	2025-02-24 18:55:00	8.472793e+01	0.000000	7.442939e+10	0.089578	5.
75%	101.580000	1.879025e+02	2.058960e+03	2025-04-02 18:22:00	9.337278e+03	5.792797	8.335543e+10	0.393811	8.
max	109004.000000	1.582244e+07	3.921431e+06	2025-05-01 12:13:00	3.050948e+07	135329.090100	9.014923e+10	837.471593	1.
std	29447.654868	1.042729e+05	3.657514e+04	NaN	6.738074e+05	919.164828	1.835753e+10	6.758854	3.

```
#Market Sentiment Distribution
plt.figure(figsize=(8,5))

sns.countplot(
    data=merged_df,
    x='classification',
    order=merged_df['classification'].value_counts().index
)

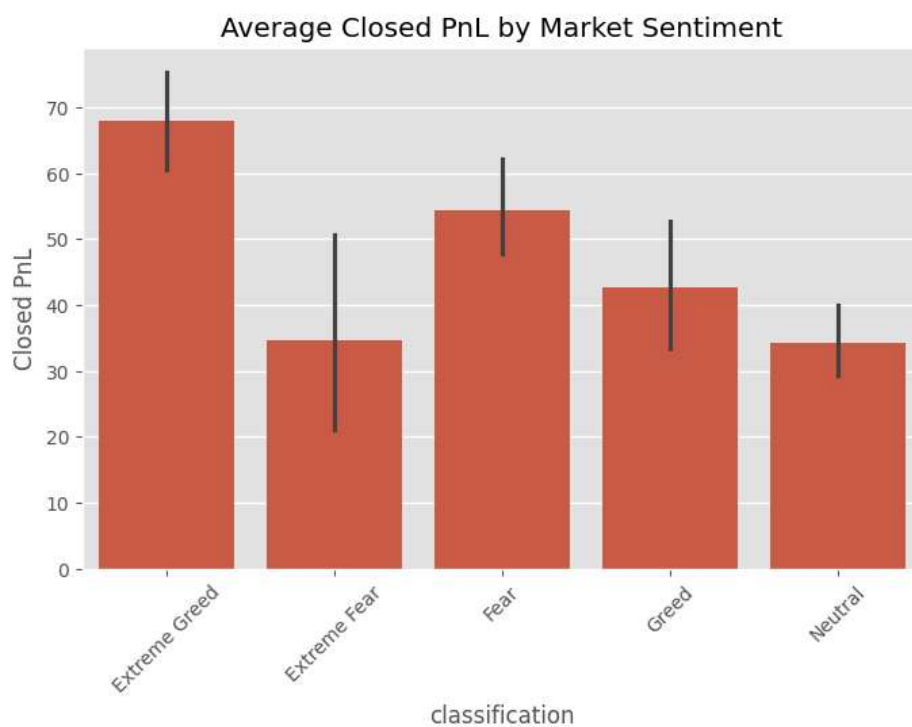
plt.xticks(rotation=45)
plt.title("Market Sentiment Distribution")
plt.show()
```



```
#Profit by Market Sentiment
plt.figure(figsize=(8,5))

sns.barplot(
    data=merged_df,
    x='classification',
    y='Closed PnL',
    estimator='mean'
)

plt.xticks(rotation=45)
plt.title("Average Closed PnL by Market Sentiment")
plt.show()
```



```
#Buy vs Sell Performance
plt.figure(figsize=(6,5))

sns.boxplot(
    data=merged_df,
    x='Side',
    y='Closed PnL'
)

plt.title("Closed PnL by Trade Side")
plt.show()
```



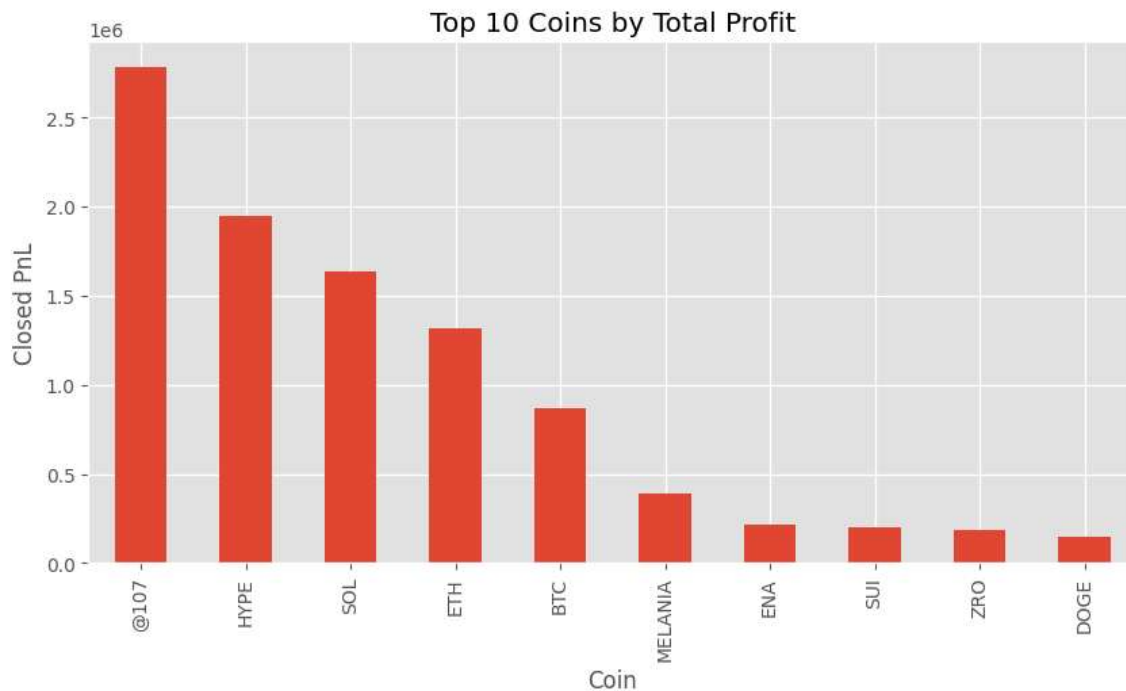
```
#Top 10 Coins by Profit
top_coin = (
    merged_df.groupby('Coin')['Closed PnL']
    .sum()
    .sort_values(ascending=False)
    .head(10)
)

plt.figure(figsize=(10,5))

top_coin.plot(kind='bar')

plt.title("Top 10 Coins by Total Profit")
plt.ylabel("Closed PnL")

plt.show()
```



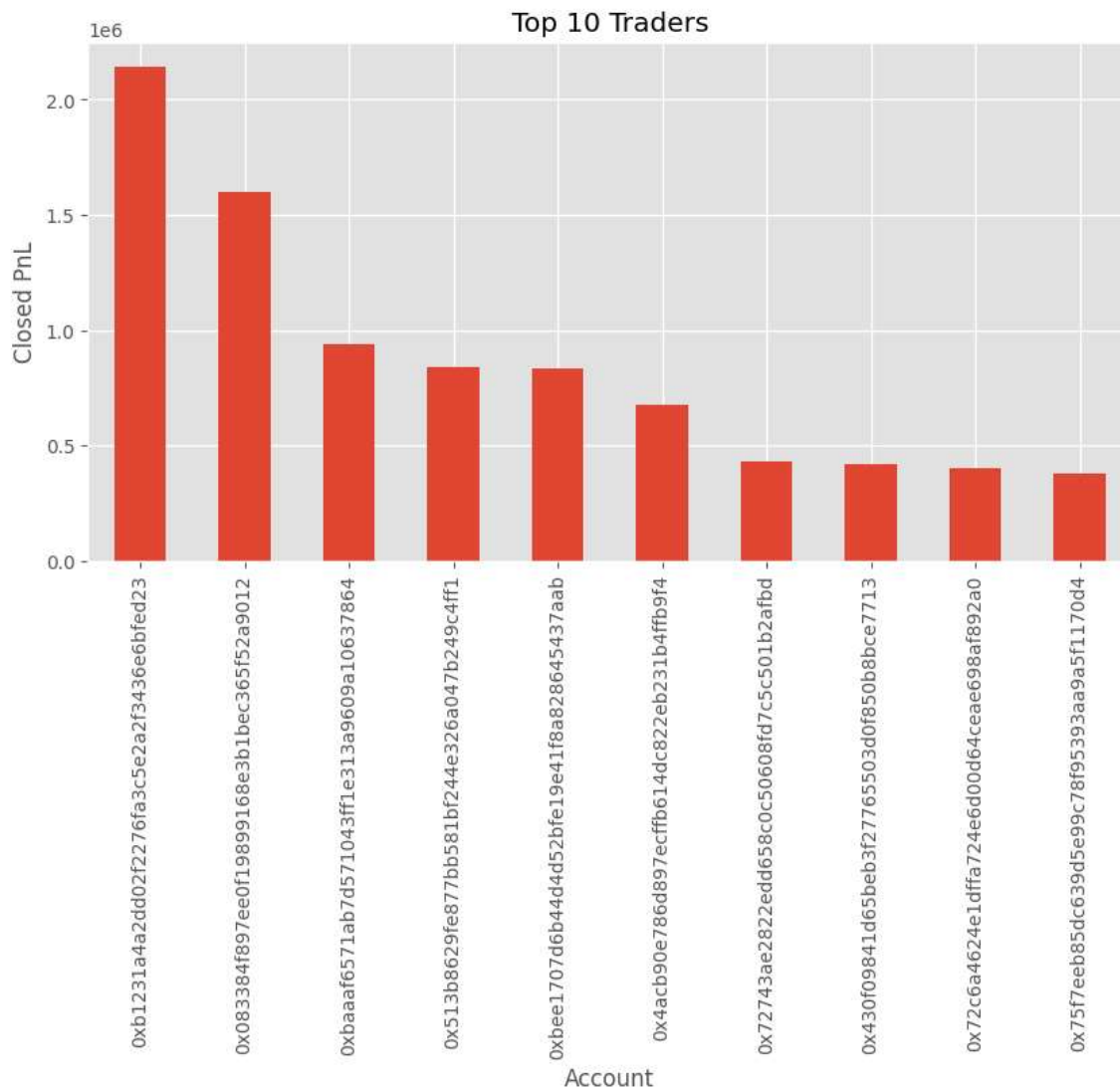
```
#Top 10 Traders
top_traders = (
    merged_df.groupby('Account')['Closed PnL']
    .sum()
    .sort_values(ascending=False)
    .head(10)
)

plt.figure(figsize=(10,5))

top_traders.plot(kind='bar')

plt.title("Top 10 Traders")
plt.ylabel("Closed PnL")

plt.show()
```



```

top_traders = (
    merged_df.groupby('Account')['Closed PnL']
    .sum()
    .sort_values(ascending=False)
    .head(10)
)

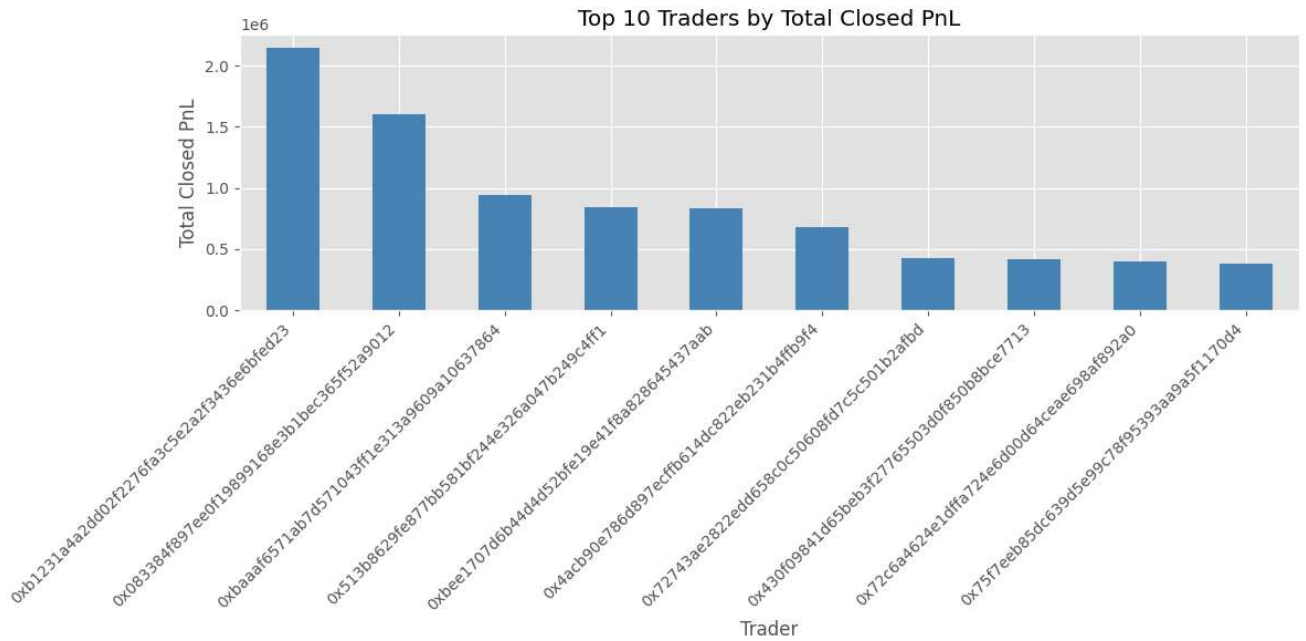
plt.figure(figsize=(12,6))

top_traders.plot(kind='bar', color='steelblue')

plt.title("Top 10 Traders by Total Closed PnL")
plt.xlabel("Trader")
plt.ylabel("Total Closed PnL")
plt.xticks(rotation=45, ha='right')

plt.tight_layout()
plt.show()

```

Profit by Market Sentiment

```
plt.figure(figsize=(10,6))

sns.barplot(
    data=merged_df,
    x='classification',
    y='Closed PnL',
    estimator='mean'
)

plt.title("Average Closed PnL by Market Sentiment")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```



Distribution of Market Sentiment

```
plt.figure(figsize=(10,6))

sns.countplot(
    data=merged_df,
    x='classification',
    order=merged_df['classification'].value_counts().index
)

plt.title("Distribution of Market Sentiment")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```

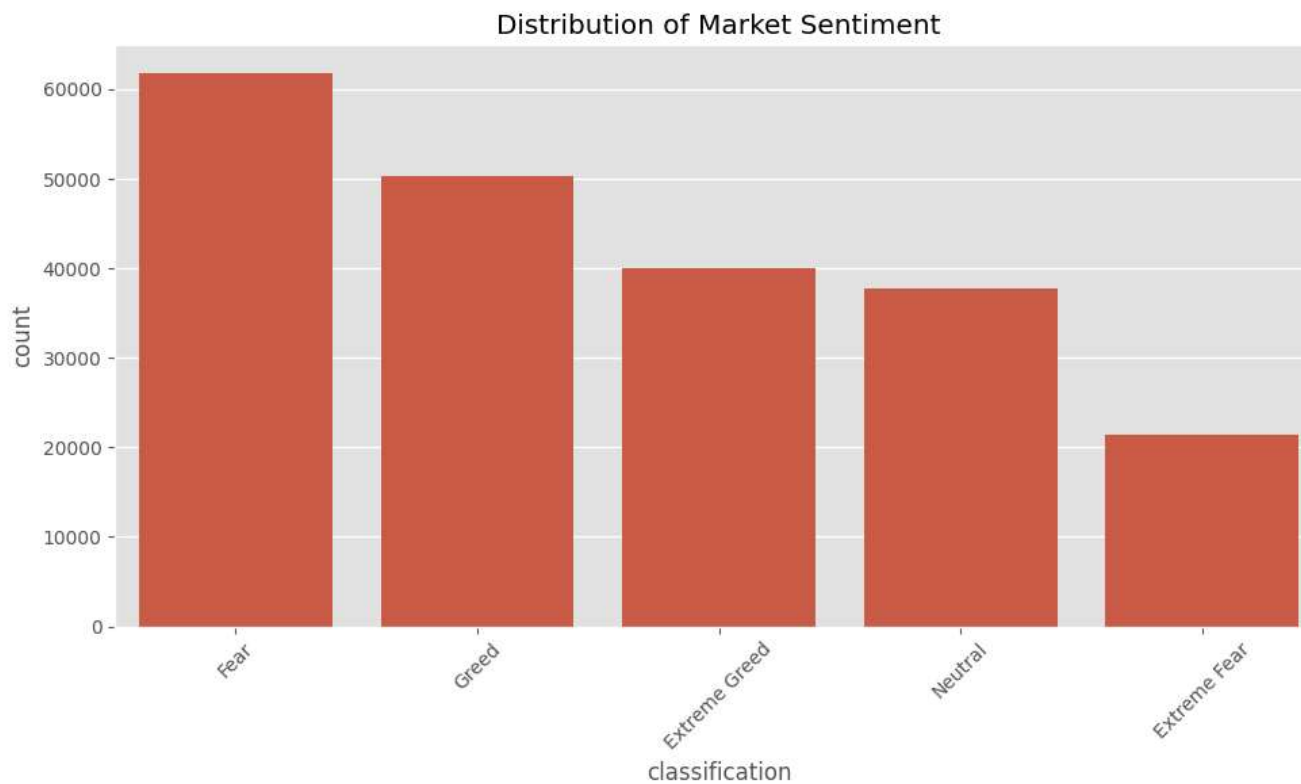


Buy vs Sell Performance

```
plt.figure(figsize=(10,6))

sns.countplot(
    data=merged_df,
    x='classification',
    order=merged_df['classification'].value_counts().index
)

plt.title("Distribution of Market Sentiment")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```



Top 10 Coins by Profit

```
top_coin = (
    merged_df.groupby('Coin')['Closed PnL']
    .sum()
    .sort_values(ascending=False)
    .head(10)
)

plt.figure(figsize=(10,5))

top_coin.plot(kind='bar')

plt.title("Top 10 Coins by Total Closed PnL")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```

Top 10 Coins by Total Closed PnL

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Market Sentiment Distribution

2.5 -

```
plt.figure(figsize=(10,5))

sns.countplot(
    data=merged_df,
    x='classification',
    order=merged_df['classification'].value_counts().index
)

plt.title("Distribution of Market Sentiment")
plt.xlabel("Market Sentiment")
plt.ylabel("Number of Trades")
plt.xticks(rotation=45)
```